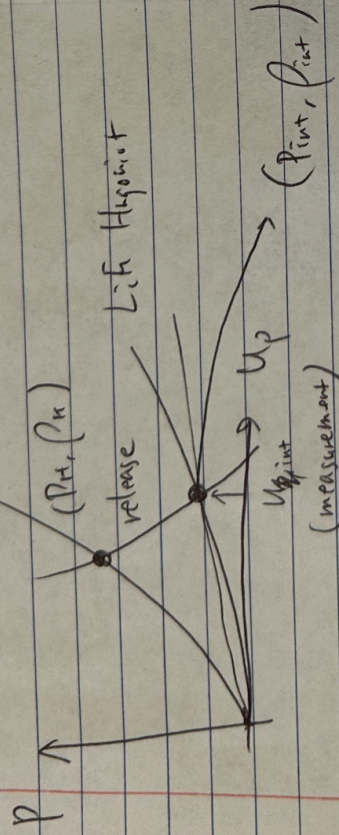


①

$$U_{\text{int}} = U_H + \int_{P_{\text{int}}}^{P_H} \frac{dP}{\rho C_s}$$

ρ_H Hugoniot



int: interface

interface particle velocity

ρ_H Hugoniot

H: Hugoniot

$$U_{\text{int}} = U_H + \int_{P_{\text{int}}}^{P_H} \frac{dP}{\rho C_s} \quad \text{release path}$$

(see like Stewart 2020, Eq 4)

$$\text{Along release, } P = P_H \left(\frac{\rho}{\rho_H} \right)^{\frac{1}{\gamma}}, \quad C = C_H \left(\frac{\rho}{\rho_H} \right)^{\frac{1}{\gamma}}$$

$$\text{Based on } C_s \text{ definition, } C_s^2 = \left(\frac{\partial P}{\partial \rho} \right) = \frac{\partial P}{\partial \rho}$$

$$C_s = \sqrt{\frac{\partial P}{\partial \rho}}$$

②

$$\frac{1}{\rho C_s} = \frac{1}{\rho \sqrt{\frac{\partial P}{\partial \rho}}} = \frac{1}{\sqrt{\partial P}} = \frac{1}{\sqrt{\partial P \left(C_H \left(\frac{\rho}{\rho_H} \right)^{\frac{1}{\gamma}} \right)^{\frac{1}{\gamma}}}} = \sqrt{\frac{\rho_H^{\frac{1}{\gamma}}}{\partial P}} \cdot \frac{1}{\rho^{\frac{1}{\gamma}}}$$

$$\int_{P_{\text{int}}}^{P_H} P \, dP = \left[\frac{2P}{2-1} P^{\frac{2-1}{2}} \right]_{P_{\text{int}}}^{P_H}$$

$$= \frac{2P}{2-1} P_H^{\frac{2-1}{2}} \left(1 - \left(\frac{P_{\text{int}}}{P_H} \right)^{\frac{2-1}{2}} \right)$$

$$\int_{P_{\text{int}}}^{P_H} \frac{dP}{\rho C_s} = \sqrt{\frac{P_H^{\frac{1}{\gamma}}}{\partial P}} \frac{2P}{2-1} P_H^{\frac{2-1}{2}} \left(1 - \left(\frac{P_{\text{int}}}{P_H} \right)^{\frac{2-1}{2}} \right)$$

$$= \frac{P_H^{\frac{1}{\gamma}}}{\partial P} \frac{2P}{2-1} P_H^{\frac{2-1}{2}} \left(1 - \left(\frac{P_{\text{int}}}{P_H} \right)^{\frac{2-1}{2}} \right)$$

$$= \frac{2}{2-1} \sqrt{\frac{\partial P_H}{\rho_H}} \left(1 - \left(\frac{P_{\text{int}}}{P_H} \right)^{\frac{2-1}{2}} \right)$$

Thus

$$U_{\text{int}} = U_H + \frac{2}{2-1} \sqrt{\frac{\partial P_H}{\rho_H}} \left(1 - \left(\frac{P_{\text{int}}}{P_H} \right)^{\frac{2-1}{2}} \right)$$